Milestone2 Progress Report

MedAssist AI: Medical Symptom Analysis & Disease Prediction System

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Milestone2 Progress Report

Table of Contents

Milestone2 Progress Report

Project Title: MedAssistAI – Medical Symptom Analysis & Disease Prediction System

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1. Objective

The objective of Milestone 2 was to develop the backend foundation for the Risk Assessment module. The work focused on implementing FastAPI APIs, request validation, risk score calculation, disease severity classification, health flag detection, and Swagger API testing for seamless integration with the Disease Prediction module.

2. Work Completed

2.1 Backend Setup

Configured the FastAPI backend project and organized the project structure for the Risk Assessment module.

2.2 Request Validation

Designed request models using Pydantic to validate patient information such as symptoms, age, medical conditions, predicted disease, prediction confidence, blood pressure, and blood sugar level.

2.3 Risk Assessment API

Developed the POST /risk-assessment API endpoint to receive patient data and process risk assessment requests.

2.4 Business Logic

Implemented the core risk assessment logic in risk_service.py to calculate the patient's overall risk score based on:

- Age
- Medical Conditions
- Predicted Disease
- Prediction Confidence
- Blood Pressure
- Blood Sugar Level

Milestone2 Progress Report

2.5 Risk Classification

Categorized patients into appropriate risk levels based on the calculated risk score.

2.6 Disease Severity

Implemented disease severity classification based on the predicted disease.

2.7 Health Flag Detection

Added automatic detection of:

- High Blood Pressure
- High Blood Sugar

2.8 Recommendations

Generated healthcare recommendations according to the patient's calculated risk level.

2.9 API Testing

Successfully tested the Risk Assessment API using Swagger UI with different patient scenarios and validated the API responses.

3. Technologies Used

Backend:

- FastAPI,
- Python

Validation:

- Pydantic

Testing:

- Swagger UI

Version Control:

- Git & GitHub

4. Current Module Status

Module	**Status**
FastAPI Backend	Completed
Request Validation	Completed

Risk Assessment API	Completed
Risk Score Calculation	Completed
Disease Severity Classification	Completed
Risk Level Classification	Completed
Health Flag Detection	Completed
Recommendation Generation	Completed
Swagger API Testing	Completed

5. Challenges Faced

The main challenges included designing an effective risk scoring logic, validating patient input, integrating disease prediction inputs with the risk assessment workflow, and ensuring accurate API responses. These challenges were resolved through modular backend development and extensive Swagger testing.

6. Next Milestone

The next phase will focus on integrating the Risk Assessment module with the Disease Prediction module, connecting the backend APIs to the frontend, performing end-to-end testing, and deploying the complete MedAssistAI application.

7. Conclusion

Milestone 2 successfully completed the backend development of the Risk Assessment module. The module now supports validated patient input, risk score calculation, disease severity classification, health flag detection, recommendation generation, and Swagger API testing. It is ready for integration with the remaining components of the MedAssistAI system.